

Annex VI — WR Preparation Stages: Advance Preparation, Predictive Preparation, Real-Time Support, and Feedback

BEAP™ / WR Desk™ / Optirando™ Canonical Specification — standalone Annex. Version 2.3 (see §VI.9).

Document status: Normative unless explicitly marked as informative.

Applies to: WR Desk™ runtime, WR Graph™, the Night Pipeline (Annex III), the WR Preparation stages (Annex II §II.6.5), Field Device provisioning (Annex II §II.6, Annex IV §IV.5), all external retrieval performed in service of preparation, and the presentation of preparation results to the operator.

Trademark notice: Optirando™, WR Desk™, WR Graph™, BEAP™, PoAE™, PoAC™, WRVault™, and WR Code® are used throughout as product and protocol designations of the Optirando™ system.

VI.0 Status, Scope, and Relationship to the Canon

This Annex specifies the complete lifecycle of preparation in its two named stages — **WR Advance Preparation** (explicitly triggered by the operator) and **WR Predictive Preparation** (implicitly derived by the system): how an upcoming event is declared or predicted, how the system prepares for it during night and idle periods, how preparation results are presented (summary brief with per-item deep-dive expansion), how prepared content supports the operator in real time at the event (**Real-Time Support**), how the event is debriefed afterwards to iterate the WR Graph™ (**Feedback**), and how external retrieval performed in service of preparation is protected against profiling and data exfiltration.

Annex II §II.6.5 defines the preparation trigger model and its provisioning classification; it remains authoritative for those points, and this Annex is its lifecycle elaboration. All machinery of Annex III applies unchanged and is reused, not duplicated: mutation classes, Consent Records, evidence events, the Morning Digest with its budget and meta-feedback, the escalation ladder, and the Cloud Abstraction Boundary. The non-pre-satisfaction rule of Annex II §II.2.5 applies to every artifact this Annex produces. Nothing in this Annex weakens any enforcement mechanic of Annexes I–V.

VI.1 Motivation (Informative)

Interaction quality on constrained hardware, in constrained connectivity, at constrained time is determined largely before the interaction begins. Where the future is predictable — a flight tomorrow morning, a maintenance round on Friday, a recurring weekly task — time and storage can be spent in advance so that the moment itself requires little computation, little bandwidth, and no improvisation: the relevant context is assembled, the right automations staged, the right competence loaded, the open questions already answered or already asked.

The lifecycle deliberately emulates the human brain in its optimal constellation. A well-functioning mind does exactly this four-phase loop: it consolidates during sleep (the Night Pipeline preparing

while the system is idle), it rehearses deliberately when a person knows what is coming — thinking through tomorrow's flight the evening before (WR Advance Preparation, explicitly triggered), it anticipates subconsciously from experience and current preoccupation — a topic one has been engaging with surfaces its own next steps (WR Predictive Preparation, implicitly derived), and it reflects after the event, keeping what worked and discarding what did not (the Feedback stage, instrumented by the Debrief). The architecture does not claim neuroscientific fidelity; it claims the functional pattern — preparation during rest, priming before action, fast recall at the moment, consolidation afterwards — implemented as a governed loop rather than a cache heuristic: the operator (or the system, with consent) declares what is coming; the Night Pipeline prepares; the operator reviews a brief and deepens what matters; the event is supported in real time from prepared material; and the debrief afterwards teaches the graph what the preparation was worth. The loop closes on itself: every event makes the next preparation better — the system, like a mind, gets better at preparing the more it prepares.

VI.2 Definitions

WR Advance Preparation (explicit stage). Preparation for a Declared Event, triggered explicitly by the operator — spoken, typed, or selected (VI.3).

WR Predictive Preparation (implicit stage). Preparation proposed by the system, derived from the WR Graph™, the active context, and memory — surfaced as a queue, never executed before acceptance (VI.4).

Declared Event. An upcoming context explicitly declared by the operator through a preparation directive — "Prepare for..." / "Prepare me for...", or its equivalent in any supported language — followed by the event: "Prepare for tomorrow's 09:40 flight from the airport"; "Prepare for Friday's maintenance round on line 3." The directive is the trigger; the remainder is parsed into the event descriptor: subject, time window, location or context region where applicable, and target devices.

Preparation Consent. The scoped Consent Record created by an event declaration: it covers preparation-package generation (Class 2) and a declared external-retrieval scope for this event (VI.3.1). It is bounded by the event descriptor and expires with the event unless renewed.

Preparation Run. The Night-Pipeline execution that prepares for one or more events (VI.3.2).

Preparation Brief. The presentation of a Preparation Run's results: a bounded summary with per-item deep-dive expansion (VI.3.3).

Deep-Dive Expansion. The operator-triggered deepening of a single brief item into a focused follow-up preparation (VI.3.4).

Preparation Queue. The ranked set of predicted preparation proposals presented in the Morning Digest (VI.4).

Event Session. The runtime session at the event itself, operating over prepared content (VI.5).

Debrief. The post-event feedback pass rating the preparation; the instrument of the Feedback stage (VI.6).

Query Obfuscation / Query Distribution. The retrieval-privacy mechanisms of VI.7.



VI.2.1 Ok Opty! — Conversational Orchestrator Interface and Wake Phrase

Ok Opty! is the conversational interaction surface and supported wake phrase of the Optirando™ Orchestrator. It provides a natural-language entry point to Orchestrator functions exposed by the current deployment.

Where wake-phrase activation is enabled, uttering “Ok Opty!” activates the conversational interface and opens an operator interaction session. The wake phrase itself does not select, authorize, or execute an Orchestrator function. The operator’s subsequent utterance is interpreted against the functions supported by the deployment and routed to the applicable canonical interface.

Ok Opty! is therefore not a separate control plane, execution mechanism, consent mechanism, or authorization path. It is an interaction layer over existing Orchestrator capabilities. A supported function invoked through Ok Opty! remains governed by the same trigger, scope, consent, provenance, capability, mutation-class, Finalizer, PoAE™, and other authorization rules that would apply if the same function had been invoked through another supported interface.

The interface MAY be used to invoke, configure, inspect, continue, or query supported Orchestrator functions. The exact functions available through Ok Opty! are deployment-dependent. WR Advance Preparation is one example; the Ok Opty! interface itself is not specific to preparation.

Companion-device interaction and capture

An enrolled smartphone or other companion device MAY serve as an interaction, capture, and presentation endpoint for Ok Opty!. Its microphone MAY carry spoken interaction, its camera MAY capture operator-provided material, and its display MAY present confirmations, intermediate states, Preparation Briefs, or other Orchestrator results.

Interaction placement, capture placement, compute placement, and presentation placement are independent.

The device on which the operator speaks or captures material therefore need not be the device on which the invoked function performs its primary computation. A smartphone MAY provide microphone, camera, sensors, and display while a more capable Host performs the computational workload using its available GPU, CPU, RAM, storage, local models, admitted adapters, and other deployment resources.

Captured material remains subject to the ordinary rules of the invoked function. A camera capture does not create an independent authority or trust boundary.

Documents, images, tickets, confirmations, or other operator-provided material enter the applicable workflow as user-provided Sources where the governing specification permits them and remain subject to the applicable reconstruction, provenance, consent, classification, and admission rules.

Compute placement likewise does not change authorization semantics. Moving computation from the companion device to the Host does not expand the operator's consent, retrieval scope, capabilities, or authority.

Example — triggering WR Advance Preparation through Ok Opty!

An operator preparing for a journey MAY state:

“Ok Opty! Tomorrow at 12:00 I am flying to Barcelona. Prepare accordingly. I want to review it in eight hours.”

In this example, the interaction consists of several distinct architectural steps.

1. Wake-phrase activation.

“Ok Opty!” activates the conversational Orchestrator interface. It does not itself initiate a Preparation Run.

2. Intent resolution.

The remainder of the utterance is resolved as a request for WR Advance Preparation and routed to the Advance Preparation interface defined by §VI.3.

3. Local speech processing.

The spoken declaration is transcribed within the deployment by the applicable local speech-to-text pipeline. Because the declaration describes future operator intent, its transcription remains subject to the local-processing requirement of §VI.3.1 and SHALL NOT depend on an external speech-transcription service.

4. Structured event declaration.

The resulting utterance is parsed into the applicable structured event descriptor. For this example, the descriptor MAY contain the travel subject, the declared departure

time, the destination, the applicable context or location information, the target devices, and the requested time by which the operator wishes to review the result.

The parsed descriptor is presented back to the operator for confirmation. The spoken utterance itself is not the deterministic preparation trigger. Only the confirmed structured descriptor becomes the Declared Event and binds the resulting Preparation Consent as specified in §VI.3.1.

5. Companion capture.

During the same Ok Opty! interaction, the operator MAY use the smartphone camera to capture a flight ticket, booking confirmation, itinerary, boarding document, or other relevant material.

Such material enters the preparation context as a user-provided Source under the applicable Preparation Consent. Where the material is an image, document, or other reconstructable media object, the reconstruction rules of Annex V apply unchanged.

The smartphone therefore acts as the acquisition endpoint; it does not need to perform the resulting preparation workload locally.

6. Host-side Preparation Run.

Once the declaration is confirmed, the Preparation Run MAY execute predominantly or entirely on a more capable Host.

The Host MAY perform the ordinary stages defined by §VI.3.2, including context assembly, event-scoped gap analysis, retrieval within the confirmed scope, candidate-artifact generation, verification, competence or adapter resolution, and packaging.

Where the Host contains substantially greater compute capability than the companion device, the scheduler MAY exploit that capability for preparation. This MAY include larger local models, longer inference runs, iterative analysis, multi-stage verification, or other resource-intensive preparation strategies permitted by the applicable preparation policy.

The smartphone may remain only the microphone, camera, interaction, and presentation endpoint while the Host performs the computational work.

7. Requested review deadline.

The phrase “I want to review it in eight hours” MAY be interpreted as a requested preparation deadline.

Where applicable, the scheduler MAY associate that deadline with a bounded preparation window under §VI.3.2.1. The remaining interval becomes an execution resource: the scheduler MAY trade immediate response latency for model capability,

analytical depth, verification effort, iterative refinement, or result quality, provided that the preparation is expected to complete before the requested review time.

The requested deadline changes the available execution horizon; it does not expand the Preparation Consent, retrieval scope, mutation class, provenance requirements, or authorization boundaries.

8. Result presentation.

On completion, the resulting Preparation Brief MAY be presented on the smartphone, on the Host, on another declared device, or through another permitted Orchestrator surface.

The placement of the result therefore remains independent of where the declaration was spoken and where the preparation computation occurred.

At the later event, Real-Time Support MAY use host-served prepared content while connectivity is available or a previously provisioned, sealed, lease-bound package where the target device must operate independently.

Governing rule

Ok Opty! makes supported Orchestrator functions conversational; it does not make them less governed.

Wake-phrase activation, natural-language interpretation, camera capture, companion-device interaction, and Host-side computation SHALL NOT pre-satisfy any consequential authorization requirement. Any operation that reaches an authorization boundary remains subject to the ordinary consent, capability, Finalizer, and PoAE™ rules at the point of consequence.

VI.3 WR Advance Preparation — the Explicit Trigger

VI.3.1 Declaration (Normative)

An event declaration is a deterministic preparation trigger (Annex II §II.6.5, trigger 1) available on demand, independent of the Night-Pipeline schedule.

The explicit trigger is a natural-language declaration in any supported language, spoken or typed — "Prepare the flight for tomorrow; here are the exact details." Spoken declarations SHALL be transcribed by a local speech-to-text model (a Whisper-class model executed within the deployment's local pipeline): the utterance describes the operator's future intent — precisely the data class VI.7 exists to protect — and SHALL therefore never be transcribed by an external service. The local pipeline parses the transcription or typed input into the structured event descriptor (subject, time window, location where stated, target devices).

A declaration MAY include materials handed over with it — "here are the exact details": booking confirmations, documents, itineraries. Such materials enter as user-provided Sources (Annex III

§3.3, acquisition: user-provided) under the Preparation Consent, and are processed per the reconstruction rules of Annex V where they are media or documents.

The parsed descriptor SHALL be presented back to the operator for confirmation before it takes effect: the Preparation Consent binds to the confirmed structured descriptor, never to the raw utterance, so that a misparse — in any language — cannot silently create a wrong scope. The confirmed descriptor is the deterministic trigger; the language interface is merely its most convenient entry.

A declaration SHALL record the event descriptor and SHALL create a Preparation Consent whose scope covers:

- assembly of preparation packages from already-admitted graph content (Class 2, per Annex III §4.5), and
- a declared external-retrieval scope for this event: the subject areas for which the system may retrieve public material, subject in full to the retrieval-privacy rules of VI.7 and the boundary discipline of Annex III §7.

Anything beyond this scope — ingestion of sources outside the declared subject areas, acquisition of external artifacts, adapter training — is not covered by the declaration and SHALL be surfaced as an ordinary Class-3 proposal in the Preparation Brief. The declaration authorizes preparation; it never authorizes execution: no PoAE™ tier is pre-satisfied by any declared event (Annex II §II.2.5).

VI.3.2 Preparation Run (Normative)

During night or idle periods — or on demand where the declaration is time-critical — the Preparation Run executes for each pending Declared Event:

1. **Context assembly** (Class 1–2): the relevant graph regions, episodic history, applicable templates and automation paths, and applicable competence bindings are resolved against the event descriptor.
2. **Event-scoped gap analysis** (produces proposals): the assembled context is tested against the event's foreseeable needs — the gap typology and evidence requirements of Annex III §5 apply unchanged; a gap without evidence of relevance to the event is not surfaced.
3. **Retrieval within scope** (consent-scoped; privacy per VI.7): public material within the declared retrieval scope is retrieved and staged; ingestion into the WR Graph™ classifies per Annex III (new sources: Class 3, presented in the Brief with provenance plans — staged material not accepted in the Brief is discarded, not silently retained).
4. **Candidate artifacts** (classified per Annex III): prepared context packages; staged templates and automation fragments; LoRA-class adapter suggestions — activation of already-admitted adapters relevant to the event (Class 2 re-ranking) or acquisition/training suggestions where the ladder justifies them (Class 3, escalation ladder of Annex III §6.2 unchanged, external acquisition before training); PoAC™ extension proposals — new sources, elicitation requests, WCR recording proposals arising from event-scoped gaps (all Class 3).
5. **Packaging** (Class 2): accepted-by-default content (already-admitted, non-confidential) is sealed, version-stamped, and lease-bound per Annex II §II.6.2; provisioning to the declared devices follows the classes of §II.6.2 and, in field deployments, Annex IV §IV.5.

A Preparation Run is interruptible and resumable; each stage is independently useful.

VI.3.2.1 Night Preparation Window (Normative)

Night Preparation Window. A Declared Event MAY be assigned an explicit Night Preparation Window during setup or confirmation. A Night Preparation Window is distinct from ordinary idle execution: it provides a known, bounded multi-hour execution horizon for preparation workloads whose expected benefit depends on sustained computation and that would be unsuitable for opportunistic idle scheduling. Within such a window, the remaining time until the expected use of the prepared result becomes an execution resource. The scheduler MAY therefore derive a preparation strategy from the available time budget and trade latency for model capability, analytical depth, verification effort, or result quality, provided that completion is expected before the event or other declared preparation deadline. This MAY include multi-stage or multi-model analysis, iterative refinement and verification, accelerator-memory offloading, CPU/RAM/storage-assisted inference, layer-wise or expert-wise model streaming, distributed inference, or equivalent techniques permitting computation that would be impractical during ordinary interactive or idle operation. Selection and prioritization MAY consider the remaining preparation time, expected time of use, task complexity, estimated quality gain, available compute, memory and storage bandwidth, energy or resource budgets, and checkpoint or resume capability. Enabling a Night Preparation Window changes only the permissible execution strategy and resource horizon; it does not expand the Preparation Consent, retrieval scope, mutation class, provenance requirements, or any authorization boundary.

VI.3.3 The Preparation Brief (Normative)

Results are presented as a Preparation Brief: a bounded summary at the top, followed by items grouped by kind (context, automations, competence, open proposals, staged retrievals). For every item the Brief SHALL show: what it is, why it was prepared (its evidence link to the event descriptor), its provenance, and its mutation class. Class-3 items are proposals in the full Annex III sense: explicit accept / reject / defer, each decision a Consent Record; rejected item types dampen future preparation of that type (§5.6 meta-feedback — the operator trains the preparer).

The Brief for Declared Events is presented on completion of the run or with the next Morning Digest, whichever the operator's configuration selects; it does not consume the Digest's proposal budget, since its scope was individually consented at declaration.

VI.3.4 Deep-Dive Expansion (Normative)

Every Brief item SHALL be expandable. Expansion has two depths:

1. **Inspection** (immediate): the item's full content, evidence chain, provenance, and dependencies — no new computation, no new retrieval.
2. **Focused follow-up preparation** (on request): the operator marks an item or sub-point for deepening ("more on this airline's rebooking process"). The follow-up is a scoped Preparation Run for that sub-point, executed on demand or queued to the next idle window. Retrieval within the original declared scope proceeds under the existing Preparation Consent; a sub-point outside that scope requires an explicit scope extension (a new, narrower

Preparation Consent). Expansion results append to the Brief as nested items under the same rules.

Expansion depth is unbounded in principle and bounded in practice by consent scope, idle budget, and the Digest discipline; expansion never bypasses a class boundary — deepening a proposal does not accept it.

VI.3.4 Controlled-Renderer Interactive Artifacts (Normative)

Prepared interactive elements constitute an additional artifact class alongside generated HTML layouts, infographics, and image-based presentation elements. A Preparation Run MAY generate a bounded local mini-application where sliders, typed inputs, selectors, or other simple controls materially improve the operator's understanding or ability to compare alternatives. Such an artifact SHALL execute only inside the controlled WR renderer: it is treated as a sealed local page, resolved deterministically under the same renderer-governed execution model used for locally replayed WCR-recorded interactions with external websites. The renderer, not the generating model, forms the security boundary; it SHALL deny undeclared network access, unrestricted persistence, operating-system access, dynamic code loading, unauthorized capabilities, and any direct consequential effect. Generated interactivity therefore extends the existing production of prepared layouts rather than introducing a general-purpose application runtime. Artifacts MAY enrich the WR Advance Preparation view, an accepted WR Predictive Preparation result, or the Real-Time Support view composed from prepared and live context. Where an artifact proves useful through Debrief verdicts and preparation telemetry, the next Night Pipeline MAY abstract, annotate, deduplicate, version, and propose it for admission into an internal reusable library. The annotated library is thus a long-term target architecture and a governed product of actual use, not a complete prerequisite for initial deployment; over time, successful event-scoped artifacts become reusable interaction patterns that reduce repeated generation and improve deterministic session composition. The artifact itself SHALL remain non-authoritative: any consequential action derived from its output continues through the ordinary capability, consent, Finalizer, and PoAE™ boundaries. This extends the Annex's existing candidate-artifact, deterministic composition, and feedback-iteration mechanisms without weakening them.

VI.4 WR Predictive Preparation — the Implicit Trigger (Normative)

The implicit trigger is the operator's own engagement. Where enabled by configuration — the opt-in discipline of Annex III §5.7 applies identically — the system derives predicted events from the WR Graph™, the active context, and memory: the graph regions the operator is currently working in, the topics of active and recent sessions, recurring contexts, calendar-like periodicities, unfinished threads ("where work could continue"), and approaching known dates. When the operator engages with a subject, the system MAY infer the forthcoming need — what could be relevant next, where the work is likely to continue. Detection operates exclusively over locally held graph, context, and episodic evidence and is Class 1.

Predicted preparations are never executed on prediction alone. They are presented as the Preparation Queue inside the Morning Digest — ranked by expected utility × evidence strength,

within the Digest's hard budget (Annex III §5.6), each entry stating what would be prepared, why the prediction was made (evidence chain), and what its execution would cost (idle time, retrieval scope, storage). Accepting a queue entry converts it into a Declared Event with a Preparation Consent scoped exactly to the entry; rejecting or ignoring it is meta-feedback that dampens the predictor. No external retrieval, no package generation, and no provisioning occurs for a predicted event before acceptance.

Cross-context analysis and Insight Artifacts (reserved). WR Predictive Preparation MAY analyze across multiple contexts at three levels: scoped (analysis strictly within one context), cross-context (multiple contexts read, producing exclusively referenced intermediate results), and meta-context (priorities, patterns, summaries, recommendations, and cross-cutting insights — never operative project data). Cross-context and meta-context outputs are represented as **Insight Artifacts**: graph entities that carry region *references* and derived statements, and are structurally incapable of embedding region raw content. Their derivation over admitted content is Class 1 per Annex III §4.5; admission of any resulting new knowledge into the WR Graph™ remains Class 3 unchanged; the region-isolation and transport rules of Annex IV §§IV.2, IV.5.2 are unaffected — an Insight Artifact never substitutes for, and never transports, region content. The full Insight Artifact specification is reserved for a subsequent revision; this paragraph fixes its defining constraint (references, never raw content) so that no interim implementation violates it.

VI.5 Real-Time Support (Normative)

Real-Time Support is not an independent module: it dynamically composes the outputs of WR Advance Preparation and WR Predictive Preparation together with live context into the operator-facing result at the moment of the event; the composition mechanics — including pre-generated layout candidates and deterministic selection — are specified by the Session Composition Pool of Annex VIII.

1. **Activation.** An Event Session activates through deterministic triggers only: arrival of the declared time window, resolution of the declared location or context region (including WR Code® scan where the event is anchored to one), topic resolution — a question or session concerning the prepared subject alone suffices, classified against the graph exactly as in Annex II §II.2.2, or explicit operator selection (Annex II §II.2.2). Activation loads the prepared package, applies the suggested competence bindings that were accepted, and stages the accepted automation paths.
2. **Placement.** Prepared content is served per the deployment's placement modes: host-served where connected; from the provisioned, sealed, lease-bound package on the device where offline (Annex II §II.6.2–II.6.3; Annex IV §IV.5.1). The lease semantics of Annex II §II.3.1 govern offline validity throughout the event.
3. **Prepared-first resolution.** During the event, resolution SHALL prefer prepared context; live retrieval is a fallback within policy and connectivity, subject to VI.7 where external. The operator experiences fast, context-aware interaction on whatever hardware the event allows — that is the purpose of the preparation.
4. **Divergence.** Where the event diverges from its preparation — a different terminal, a changed procedure, a question outside the prepared scope — the system degrades gracefully per Annex II §II.2.8: the absence of prepared context is surfaced as such; nothing is

- fabricated from stale preparation. Divergences are recorded as evidence events for the Debrief.
5. **Authorization.** Nothing prepared, staged, or suggested pre-satisfies any authorization requirement: every consequential step during the Event Session is gated at its applicable PoAE™ tier at the point of consequence (Annex II §II.2.5; on subordinate devices, Annex IV §IV.8.3).
 6. The Event Session, its resolutions, its divergences, and its outcomes are recorded as episodic evidence scoped to the event.

VI.6 The Feedback Stage: Post-Event Debrief (Normative)

The Debrief is the post-event instrument of the Feedback stage: it converts the operator's assessment of the preparation into the evidence that drives the next iteration.

1. After the event window closes, the system SHALL offer a Debrief: a compact review of the preparation against the actual event. Per prepared item, the operator MAY record a verdict — used and helpful / used but wrong / unused / something was missing — with optional commentary. Each verdict is a Feedback Event per the schema of Annex III §5.2, with `target_ref` = the preparation item and `project_ref` = the event.
2. **Preparation telemetry.** Independently of explicit verdicts, the system SHALL record which prepared items were actually resolved, activated, or executed during the Event Session — the preparation analog of adapter telemetry (Annex III §6.3).
3. **Iteration.** The next Night Pipeline integrates Debrief and telemetry under the standard classes: confidence updates and evidence linking (Class 1); re-ranking and supersession of preparation paths that proved stale (Class 2, lineage preserved); new knowledge arising from "something was missing" (Class 3 proposals with the Debrief as evidence chain). Item types that are repeatedly prepared but unused SHALL be dampened — over-preparation is a defect the loop corrects, not a virtue.
4. The Debrief is optional per event and never blocking; an event without a Debrief still contributes telemetry.

VI.7 Retrieval Privacy: Masking, Obfuscation, and Query Distribution

VI.7.1 Threat Statement (Informative)

Preparation queries are uniquely revealing: they describe the operator's future — locations, timings, intentions, business activity — before it happens, concentrated in a short time window, often the night before. An external provider observing them can profile plans and movements; correlated across events, they exfiltrate a life pattern. The metadata rationale of Annex III §7.3 ("usage frequencies leak business activity") applies here with sharpened force: it is not past activity that leaks, but intent.

VI.7.2 Baseline Discipline (Normative)

All external retrieval in service of preparation is subject to the Cloud Abstraction Boundary discipline of Annex III §7: only permitted outbound data classes cross; prohibited classes never cross under any configuration; retrieved material enters as Sources with full provenance and passes

the local instantiation step (§7.4). Preparation in its entirety SHALL be operable with external retrieval disabled — preparation from the graph and local corpus alone is always a conforming mode (§7.5); external retrieval is an accelerant, never a dependency.

VI.7.3 Masking (Normative)

Before any preparation query leaves the deployment, it SHALL be masked: stripped of operator identity and identity-linkable detail, generalized to the least-specific form that still retrieves the needed material, and screened by the local classifier against the prohibited classes of Annex III §7.3. Event descriptors themselves — who is going, with whom, why — SHALL NOT appear in outbound queries; only the subject matter required for retrieval may.

VI.7.4 Obfuscation Mode (Normative, Configurable)

A deployment MAY enable Obfuscation Mode for preparation retrieval. Under Obfuscation Mode, outbound retrieval SHALL additionally apply, per policy, a combination of:

- **query generalization** beyond the masking baseline (broader subjects, retrieved locally-filtered),
- **temporal decoupling** — queries scheduled so that their timing does not correlate with the event (spread across idle windows rather than concentrated the night before),
- **pattern normalization** — query volume and cadence padded toward the deployment's ordinary background, and
- **cover queries** — decoy retrievals within plausible adjacent subjects (MAY; with the cost and residual-detectability limits of VI.7.6).

Obfuscation Mode is optional by default. **In High-Assurance Mode, Obfuscation Mode SHALL be enforced for all preparation retrieval:** where High-Assurance policy does not permit its overhead, the conforming alternative is local-only preparation (VI.7.2), never un-obfuscated external retrieval. Enabling, disabling, or reconfiguring Obfuscation Mode is a policy configuration change (Class 3, Annex III §4.5).

Obfuscation Mode is not limited to retrieval: in High-Assurance Mode it extends to transport (padding, timing decoupling, cover traffic on the sealed transport paths) as specified in Annex IX; this section governs retrieval exclusively, and the two applications share one canonical concept.

VI.7.5 Query Distribution (Normative)

Where Obfuscation Mode is active and external retrieval occurs, the retrieval set SHALL be distributed: partitioned across multiple independent external providers such that no single provider observes the complete intent of any event's preparation. Partitioning SHALL be subject-disjoint where feasible (different providers see different facets, not copies); assembly of the retrieved material into event context occurs exclusively locally (Annex III §7.4). Distribution requires at least two configured, independent providers; where only one is available, the deployment SHALL either fall back to local-only preparation or obtain an explicit per-event consent acknowledging single-provider exposure — in High-Assurance Mode, only the local-only fallback is conforming.

VI.7.6 Honest Limits (Informative)

Obfuscation reduces inference; it does not eliminate it. A sufficiently distinctive subject identifies itself regardless of masking; cover queries cost bandwidth and are themselves statistically

detectable in principle; distribution protects against any single provider, not against colluding providers or an observer of the network path (transport-level protections are governed by the Canon's transport specifications and Annex IX, not by this Annex). The strongest privacy guarantee remains what the baseline already provides: local-only preparation, which this Annex keeps permanently available and which High-Assurance deployments should prefer wherever the local corpus suffices.

VI.8 Worked Example — The Airport (Informative)

The operator declares in the evening, spoken, transcribed by the local speech model: "Prepare tomorrow's flight — 09:40 from the airport, here's the booking confirmation." The attached confirmation enters as a user-provided Source; the parsed descriptor — subject, time window, location, target device: her phone — is shown back and confirmed with one tap. The declaration creates a Preparation Consent scoped to travel-procedural subjects.

Overnight, the Preparation Run assembles what the graph already holds (her airline's procedures from past trips, the standing check-in automation, the packing checklist recording), finds two event-scoped gaps (the airport changed its security-lane system; her connecting airport is new to the graph), retrieves public material for both — masked, and because Obfuscation Mode is on, distributed across two providers and spread over the night's idle windows — and stages a language-competence activation for the destination. The package is sealed to her phone with a lease until the end of the travel day.

In the morning, the Preparation Brief: a five-line summary; beneath it, the items. She deep-dives the new security-lane point, accepts the two staged sources (Class 3, one tap each), rejects a suggested lounge-finder automation (that type will be dampened), and leaves.

At the airport, the declared time window and location activate the Event Session. Offline in the terminal basement, everything answers from the provisioned package; the check-in automation runs to its authorization boundary and waits for her confirmation — the declaration prepared it, but nothing pre-satisfied the gate. One divergence — a changed boarding procedure — is surfaced as not-prepared, answered from the base model, and recorded.

That evening, the Debrief: security-lane item used and helpful; the destination-airport package unused (the connection was through a different terminal — telemetry already knew); boarding-procedure divergence becomes a Class-3 source proposal for next time. The graph iterates; the next trip's Brief is shorter and sharper.

§VI.9 Version History

Version 1.0

Date: 2026-07-09

Initial normative specification: preparation lifecycle — event declaration with scoped Preparation Consent, incl. natural-language declaration in any supported language (spoken via local Whisper-class transcription — never external — or typed), attached materials as user-provided Sources under the consent, and mandatory confirmation of the parsed descriptor before consent takes effect; the two modes explicitly named (explicit trigger; implicit trigger, derived from WR Graph™, active context, and memory); brain-emulation framing of the lifecycle (sleep consolidation, deliberate

rehearsal, subconscious anticipation, post-event reflection) in the motivation; Preparation Run stages (context assembly, event-scoped gap analysis, consent-scoped retrieval, candidate artifacts incl. LoRA-class adapter suggestions and PoAC™ extension proposals, sealed lease-bound packaging); Preparation Brief with per-item evidence, provenance, class markers, and consent decisions; Deep-Dive Expansion (inspection and focused follow-up preparation with scope discipline); the implicit stage as opt-in Preparation Queue in the Morning Digest with no pre-acceptance retrieval or generation; real-time event support (deterministic activation, placement per existing modes, prepared-first resolution, graceful divergence handling, non-pre-satisfaction); Post-Event Debrief with per-item verdicts, preparation telemetry, class-mapped iteration, and over-preparation dampening; Retrieval Privacy — masking baseline, configurable Obfuscation Mode (generalization, temporal decoupling, pattern normalization, cover queries) mandatory in High-Assurance, Query Distribution across independent providers with local-only assembly and single-provider fallback rules, honest-limits statement; worked example.

Version 2.0

Date: 2026-07-13

Canon-wide terminology migration: the "Forward Preparation" umbrella term is retired and decomposed into the WR Preparation stage vocabulary. Annex retitled. The explicit mode is renamed **WR Advance Preparation**, the implicit mode **WR Predictive Preparation**; §VI.5 retitled **Real-Time Support** (aligning with the pipeline stage; the Event Session term unchanged); §VI.6 retitled **The Feedback Stage: Post-Event Debrief**, with the Debrief identified as the Feedback stage's post-event instrument. The lifecycle-class vocabulary (declared / predictive / real-time) is retired in favor of the stage names; the Declared Event term is deliberately retained (it names the event, not a lifecycle class). §VI.9 history text of version 1.0 adjusted retroactively; no semantic change from the renames. Three additions: §VI.5 lead-in normatively anchors Real-Time Support as the composition stage — not an independent module; composes WR Advance and WR Predictive Preparation outputs with live context per the Session Composition Pool of Annex VIII. §VI.4 reserves the **Insight Artifact** for cross-context predictive analysis (scoped / cross-context / meta-context levels; region references, never region raw content; Class-1 derivation per Annex III §4.5; region-isolation rules of Annex IV unaffected; full specification deferred). §VI.7.4 extended: Obfuscation Mode generalizes from retrieval to transport in High-Assurance Mode (normative home: Annex IX); §VI.7.6 transport reference updated accordingly. Infographics VI-N1, VI-N2, and VI-N4 scheduled for regeneration under the stage vocabulary (see editorial note).

Version 2.1

Date: 2026-07-24

Added §VI.3.5, **Prepared Interactive Artifacts and Library Growth**. A Preparation Run MAY generate bounded interactive elements — including calculators, sliders, typed inputs, selectors, and comparison views — as an additional prepared-artifact class alongside HTML layouts, infographics, and image-based presentation elements. Such artifacts execute exclusively as sealed local pages within the controlled WR renderer, which forms the enforcement boundary for deterministic execution and denies undeclared network access, unrestricted persistence, operating-system access, dynamic code loading, unauthorized capabilities, and direct consequential effects. Prepared Interactive Artifacts MAY enrich WR Advance Preparation, accepted WR Predictive Preparation results, and the Real-Time Support view. Debrief verdicts and preparation telemetry MAY cause

useful artifacts to be abstracted, annotated, deduplicated, versioned, and proposed for admission into an incrementally growing internal library. The annotated library is established as a long-term target architecture and a governed product of use rather than a complete prerequisite for deployment. §VI.3.2 candidate-artifact terminology extended accordingly; no authorization boundary is weakened.

Version 2.2

Date: 2026-08-08

Added §VI.3.2.1, **Night Preparation Window**. A Declared Event MAY be assigned an explicit, bounded multi-hour preparation window distinct from ordinary opportunistic idle execution. The remaining time until expected use is established as an execution resource, allowing the scheduler to derive a preparation strategy from the available time budget and trade latency for model capability, analytical depth, verification effort, or result quality while targeting completion before the event or declared preparation deadline. The execution strategy MAY include multi-stage or multi-model analysis, iterative refinement and verification, accelerator-memory offloading, CPU/RAM/storage-assisted inference, layer-wise or expert-wise model streaming, distributed inference, and equivalent compute-intensive techniques that would be impractical during interactive or ordinary idle operation. Scheduling MAY consider remaining preparation time, expected time of use, task complexity, estimated quality gain, available compute, memory and storage bandwidth, resource or energy budgets, and checkpoint or resume capability. The Night Preparation Window changes only the permissible execution strategy and resource horizon; Preparation Consent, retrieval scope, mutation classes, provenance requirements, and authorization boundaries remain unchanged.

Version 2.3

Date: 2026-08-09

Added §VI.2.1, **Ok Opty! — Conversational Orchestrator Interface and Wake Phrase**. Ok Opty! is established as the conversational interaction surface and supported wake phrase of the Optirando™ Orchestrator rather than as a preparation-specific mechanism. Wake-phrase activation opens an operator interaction session but does not itself select, authorize, or execute a function; subsequent natural-language intent is resolved against the functions exposed by the deployment and routed to the applicable canonical interface. Invoked functions retain their existing trigger, consent, capability, provenance, mutation-class, Finalizer, PoAE™, and authorization semantics. Added the architectural separation of **interaction placement, capture placement, compute placement, and presentation placement**: an enrolled smartphone or other companion device MAY provide microphone, camera, sensors, and display while computationally intensive execution occurs on a more capable Host using its available GPU, CPU, RAM, storage, local models, adapters, and other admitted resources. Companion-device camera captures enter applicable workflows as user-provided Sources under the ordinary reconstruction, provenance, consent, classification, and admission rules. Added WR Advance Preparation as a worked application of the general Ok Opty! interface, including the example declaration “**Ok Opty! Tomorrow at 12:00 I am flying to Barcelona. Prepare accordingly. I want to review it in eight hours.**” The example distinguishes wake-phrase activation, intent resolution, local speech processing, structured event confirmation, companion capture, Host-side Preparation Run, requested review deadline, and result presentation. A declared review time MAY establish a bounded preparation deadline and be mapped to the Night Preparation Window of §VI.3.2.1, allowing the scheduler to trade latency for model capability,

analytical depth, iterative refinement, verification effort, or result quality without expanding Preparation Consent, retrieval scope, mutation classes, provenance requirements, or authorization boundaries.

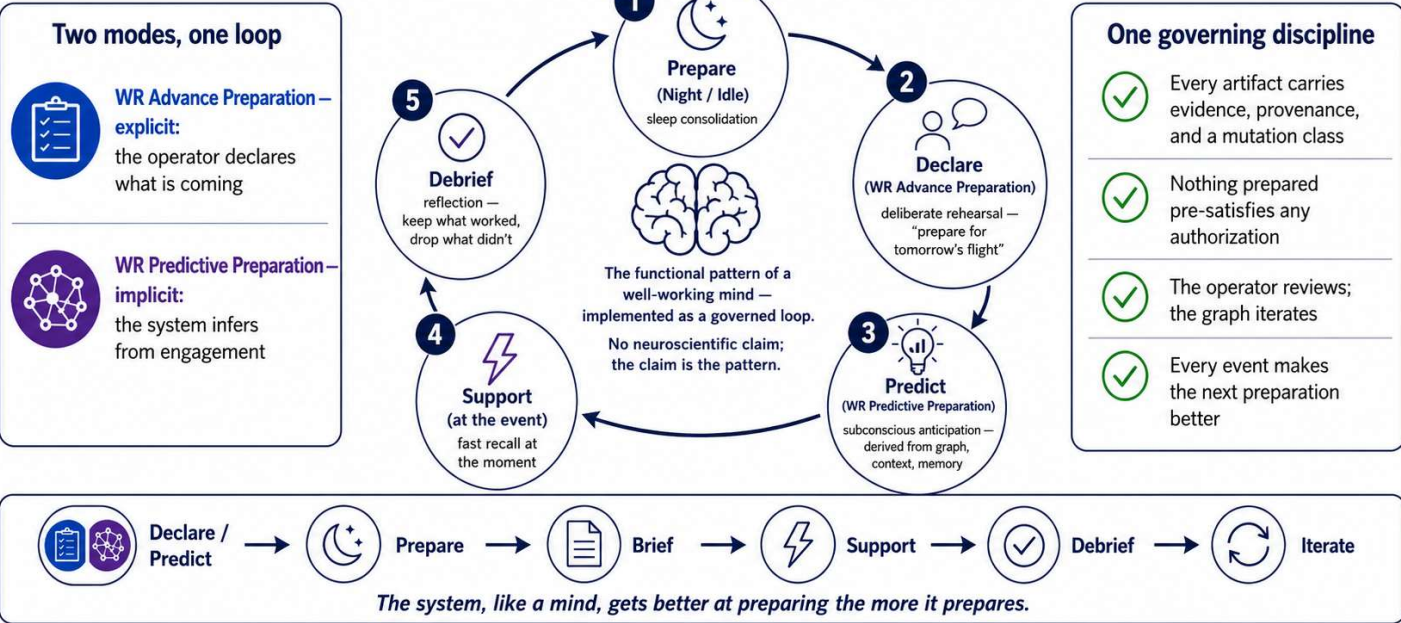
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WR Advance Preparation – A Loop That Emulates a Mind

Prepare during rest, prime before action, recall fast at the moment, consolidate afterwards

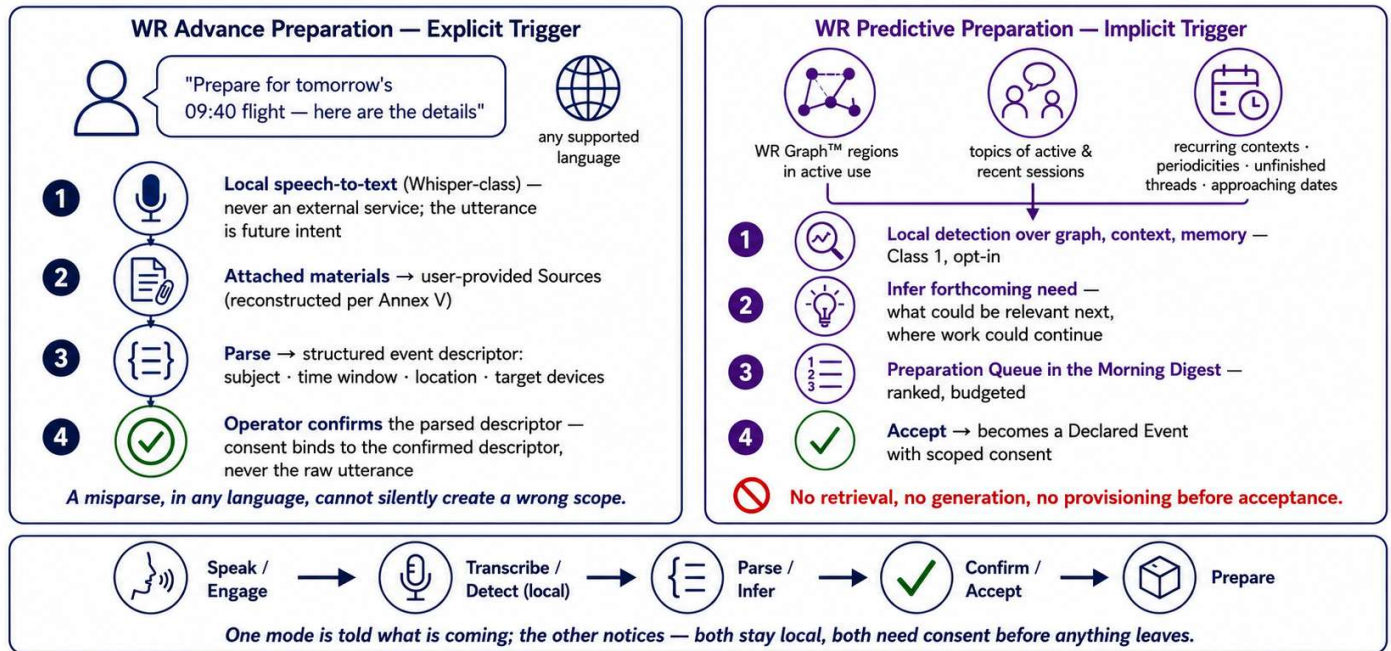
WR-AP1



Two Triggers — Spoken Intent, Silent Inference

WR-AP2

Explicit 'Prepare for...' and implicit derivation from what the operator is already doing

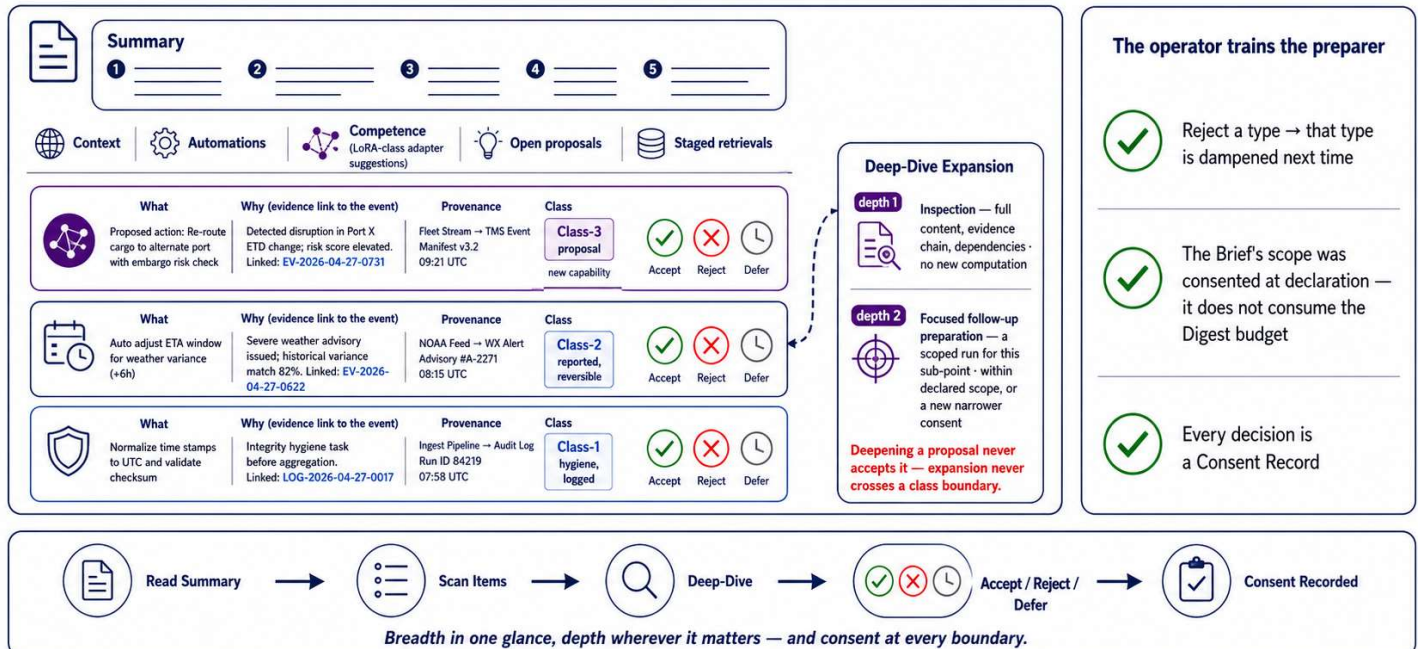


Optirando™ · WR Desk™ · Annex VI §§VI.3, VI.4 · Annex III §5.7 · 2026

The Preparation Brief — Summary First, Depth on Demand

VI-N3

Every item shows what, why, provenance, and class — and expands as far as consent allows



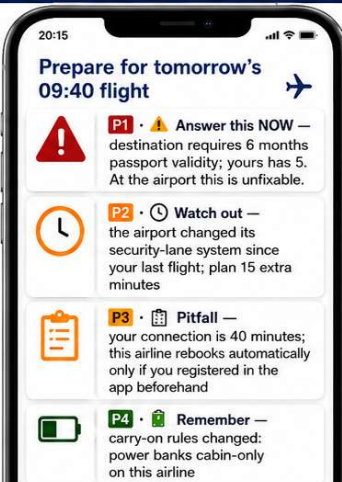
Optirando™ · WR Desk™ · PoAC™ · Annex VI §§VI.3.3–VI.3.4 · Annex III §§5.4, 5.6 · 2026

VI-N4

WR Advance Preparation — Prepared, Not Searched

Not a real-time chatbot: a priority-ordered briefing on what to watch, the pitfalls, and the questions that must be answered before — while there is still time to act.

The Evening Before — the WR Advance Brief



Prepared the evening before by WR Advance priority — derived from the declared event, your booking, your history, and event-scoped gap analysis. Some of these are only actionable now; none of them was asked.

Nothing prepared pre-satisfies authorization — the check-in automation still halts at its PoA™ tier for your confirmation.

Two kinds of questions

Real-time questions — chatbots and search do this well

How do I get from gate 3 to gate 7?

09:31 ✓

Reactive · public knowledge · instantly answerable on site

Genuinely their strength — this figure does not dispute it.

Preparation questions — WR Advance Preparation's turf

- Proactive: surfaced before they are asked
- Priority-ordered: what to watch · typical pitfalls · what to remember · what must be resolved in advance
- Grounded in your context — your booking, your history, your documents — which no public chatbot holds

The most important questions of a journey are worthless at the gate — they must be answered while there is still time to act. A question presupposes you knew to ask it; the Brief surfaces the ones you didn't.

The Loop That Learns What to Surface

Debrief — verdicts per advisory



Preparation telemetry — which advisories were acted on, recorded automatically

Iteration — “something was missing” becomes the next trip's advisory (Class 3, evidence-backed); repeatedly-unused advisory types are dampened

Every journey sharpens the next Brief.



The measure of WR Advance Preparation is not how well it answers at the gate — it is how little is left to ask there.

Optirando™ · WR Desk™ · WR Graph™ · Annex VI §§VI.3.2–VI.3.3, VI.5–VI.6 · Annex III §6.5 · 2026

VI-N5

Retrieval Privacy — Preparing Without Revealing the Future

Preparation queries describe intent before it happens — masking, obfuscation, and distribution keep it from leaking

Why this is different (the threat)

Preparation queries reveal the operator's future — locations, timings, intentions — concentrated the night before. Correlated across events, they exfiltrate a life pattern.



1. BASELINE (always)

Cloud Abstraction Boundary (Annex III §7) — only permitted data classes leave; prohibited classes never do. Local-only preparation is always a conforming mode.

2. MASKING (always)

- Strip identity and identity-linkable detail
- Generalize to the least-specific effective form
- Screen against prohibited classes

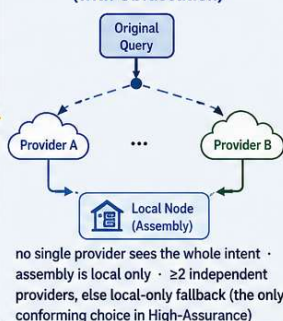
Event descriptors never leave — only the subject matter.

3. OBFUSCATION MODE (optional · mandatory in High-Assurance)

- query generalization
- temporal decoupling (spread across idle windows, not the night before)
- pattern normalization
- cover queries

Where High-Assurance won't pay the overhead → local-only, never un-obfuscated retrieval.

4. QUERY DISTRIBUTION (with Obfuscation)



HONEST LIMITS

- ✗ A distinctive subject identifies itself regardless of masking
- ✗ Cover queries cost bandwidth and are detectable in principle
- ✗ Distribution defeats one provider, not colluding providers or a network-path observer
- ✓ Strongest guarantee stays local-only preparation — always available, preferred in High-Assurance where the local corpus suffices.



Obfuscation reduces inference, it does not erase it — the surest privacy is to never ask outside at all.

Optirando™ · WR Desk™ · Annex VI §VI.7 · Annex III §7 · 2026



Ok Opty!

Conversational Orchestrator Interface & Wake Phrase

A general voice interface for the Optirando™ Orchestrator —
WR Advance Preparation is one example, not the only one.

1 What Ok Opty! is



Wake phrase

Saying "Ok Opty!" activates the conversational interface.



Not a separate control plane

It does not itself authorize, execute, or bypass governance.



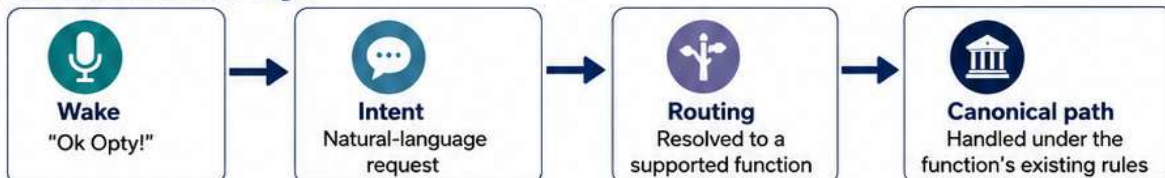
Routes to supported functions

The following utterance is interpreted and routed to the applicable Orchestrator function.



Canonical rules stay unchanged: trigger, consent, capability, provenance, Finalizer, and PoAE™ still govern the invoked function.

2 Core interaction logic



Wake phrase activates the session; it does not itself trigger execution.

3 Companion device + Host

Companion Device

- Microphone
- Camera
- Display
- Sensors



Interaction placement, capture placement, compute placement, and presentation placement are independent.

Host



- GPU
- CPU
- RAM
- Local models
- Preparation resources



Capture

Tickets, confirmations, itineraries, and other user-provided material may be captured on the phone.



Compute

The heavy workload can run on the Host while the phone remains the input and presentation endpoint.

4 Example: WR Advance Preparation

“Ok Opty! Tomorrow at 12:00 I am flying to Barcelona. Prepare accordingly. I want to review it in eight hours.”



1 Wake phrase activates session



2 Utterance resolves to WR Advance Preparation



3 Local speech pipeline transcribes the request



4 Descriptor is parsed and confirmed



5 Phone camera may capture tickets or booking details



6 Preparation Run executes on the Host



The review time may define a bounded preparation deadline.

5 Result and governance



Preparation Brief

Results may be reviewed on the phone, the Host, or another declared device.



Real-Time Support

Prepared content can later be served live or from a provisioned package.



Sources

Captured materials enter as user-provided Sources and follow ordinary provenance and admission rules.



No pre-satisfied authorization

Nothing about Ok Opty!, capture, or Host execution bypasses consent or authorization boundaries.



Ok Opty! makes supported Orchestrator functions conversational.
It does not make them less governed.